

1. Administrative & Study Identification

Full/Official Study Title: A Phase I/Ib, Open-label, Multi-center, Study of DKY709 as a Single Agent and in Combination With PDR001 in Patients With Advanced Solid Tumors **Short Title/ Acronym:** DKY709 \$\\pm\$ PDR001 Phase I/Ib in Advanced Solid Tumors **Protocol Number:** CDKY709A12101C [Novartis Internal Protocol ID] **EudraCT Number:** 2018-002580-26 **NCT Number:** NCT03891953 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** DKY709 (selective IKZF2 transcription factor degrader) and PDR001 (spartalizumab, anti-PD-1) **Phase of Development:** Phase I/Ib **Study Type:** Interventional (Primary Purpose: Treatment) **Medical Monitor:** Novartis Clinical Operations Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of DKY709 monotherapy and DKY709 in combination with PDR001, identifying the maximum tolerated dose (MTD) and/or recommended dose (RD) by assessing the incidence of dose-limiting toxicities (DLTs). **Secondary Objectives:** To characterize the pharmacokinetic (PK) parameters (AUC, Cmax, tmax, half-life) of the study drugs and to evaluate preliminary anti-tumor activity (Progression-Free Survival [PFS], Best Overall Response Rate [ORR], Duration of Response [DOR], and Time to Progression).

2.2 Background & Rationale To address the critical unmet medical need in patients with advanced solid tumors who have progressed on or are naive to prior anti-PD-1/PD-L1 therapy. While immune checkpoint inhibitors like PDR001 (spartalizumab) offer benefits, many patients exhibit primary or acquired resistance. The addition of DKY709, an investigational molecular glue/immunomodulatory agent, aims to enhance immune activation and overcome resistance in specific cohorts such as non-small cell lung cancer (NSCLC), melanoma, nasopharyngeal carcinoma (NPC), microsatellite stable colorectal cancer (mssCRC), and triple-negative breast cancer (TNBC).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose-escalation followed by dose-expansion cohorts) **Blinding:** Open-label **Randomization:** N/A (Sequential dose-escalation and parallel expansion)

3.2 Planned Enrollment & Duration Target **\$N\$:** 320 participants (Actual Enrollment: 98 participants) **Number of Sites:** Multicenter, global trial (including locations such as Massachusetts General Hospital and Dana Farber Cancer Institute in Boston, Sarah Cannon Research Institute in Nashville, as well as sites across Europe [e.g., Germany, Spain] and Asia [e.g., Japan, Hong Kong, Taiwan]) **Estimated Study Start:** 07-May-2019 **Estimated Primary Completion:** 30-Oct-2026 **Estimated Study Completion:** 31-Oct-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult patients (≥ 18 years of age) with advanced/metastatic solid tumors (specifically NSCLC and Melanoma with prior progression on anti-PD-1/PD-L1 therapies; or NPC, mssCRC, TNBC naive to anti-PD-1/PD-L1 therapies) who have progressed despite standard therapy or for whom no effective standard therapy is available. 2. ECOG Performance Status ≤ 1 . 3. Must have a site of disease amenable to core needle biopsy and be willing to undergo a tumor biopsy at baseline and during therapy. 4. NSCLC cohort patients must have historic documentation of PD-L1 $\geq 1\%$ and progressive disease after having experienced at least 4 months of investigator-assessed disease stability or response on prior anti-PD-L1-containing therapy.

4.2 Exclusion Criteria 1. Presence of symptomatic central nervous system (CNS) metastases requiring local CNS-directed therapy or increasing doses of corticosteroids within 2 weeks prior to study entry. 2. History of severe hypersensitivity reactions to any ingredient of the study drugs or other monoclonal antibodies. 3. Clinically significant cardiac disease or out-of-range laboratory values (e.g., AST $> 3 \times$ ULN, ANC $< 1.0 \times 10^9/L$, Platelet count $< 75 \times 10^9/L$, Hemoglobin < 8 g/dL).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: DKY709 administered orally at assigned dose-escalation levels, given as monotherapy or combined with PDR001 (spartalizumab) administered intravenously on a scheduled cycle. **Route of Administration:** Oral (DKY709) and Intravenous (PDR001). **Duration of Treatment:** Continued until disease progression, unacceptable toxicity, withdrawal of consent, or death.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications required to treat concurrent medical conditions or adverse events. **Prohibited:** Concurrent anti-cancer therapies, other investigational agents, and increasing doses of corticosteroids within 2 weeks prior to study entry (stable doses

for previously treated brain metastases may be permitted after an appropriate washout).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Tumor Biopsy, Safety Labs
Baseline	Day 0	First Dose (DKY709 $\pm$ PDR001), Baseline PK/PD Blood Draws
Interim	Cycle-dependent	Safety Labs, PK/PD blood draws, On-therapy Tumor Biopsy, Efficacy Imaging (RECIST 1.1)
End of Study	Follow-up	End of Treatment Visit, Survival Follow-up, Safety Monitoring (AE/SAE resolution)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability evaluated through the incidence and severity of Adverse Events (AEs), Serious Adverse Events (SAEs), and the incidence of Dose-Limiting Toxicities (DLTs) to determine the MTD/RD. **Measurement Method:** Evaluated using Common Terminology Criteria for Adverse Events (CTCAE) grading.

7.2 Secondary & Exploratory Endpoints - Pharmacokinetics (PK): Evaluation of AUC, C_{max}, t_{max}, concentration versus time, and half-life of the study drugs. - Efficacy parameters: Progression-Free Survival (PFS), Best Overall Response Rate (ORR), Duration of Response (DOR), and Time to Progression as assessed by RECIST v1.1.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of AEs and DLTs using CTCAE scales at each clinical visit and PK/PD assessment. **Serious Adverse Events (SAEs):** Must be reported to the sponsor's safety desk within standard 24-hour regulatory windows. **Data Monitoring Committee (DMC):** Internal Dose-Escalation Safety Review Committee (SRC) to evaluate DLTs and approve cohort progression.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who receive at least one dose of study medication) for MTD/RD determination; Efficacy Analysis Set for preliminary anti-tumor activity evaluations. **Sample Size Justification:** Approximately 320 patients, appropriately sized for a Phase I/Ib dose-escalation and subsequent parallel dose-expansion cohorts to capture safety

signals and initial efficacy bounds. **Handling of Missing Data:** Missing safety data generally not imputed. Efficacy analyses utilize standard censoring methods (e.g., Kaplan-Meier survival curves) for time-to-event endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence, accurate AE reporting, and PK/PD sample integrity. **Audit Trail:** Secure Electronic Data Capture (EDC) system with full audit trails for data entry, queries, and sign-offs.

11. Study Identifiers & Governance

Primary ID: NCT03891953 **Registry Number:** NCT03891953 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Study of Safety and Efficacy of DKY709 Alone or in Combination With PDR001 in Patients With Advanced Solid Tumors **Official Regulatory Title:** A Phase I/Ib, Open-label, Multi-center, Study of DKY709 as a Single Agent and in Combination With PDR001 in Patients With Advanced Solid Tumors

12. Clinical Framework (Oncology Specific)

Primary Condition: Advanced Solid Tumors (NSCLC, Melanoma, NPC, mssCRC, TNBC) **Diagnosis Threshold:** Histologically confirmed advanced/metastatic disease not amenable to standard curative therapies. **Medical Methodology:** Targeted Protein Degradation / Immunotherapy **Optimization Target:** Identification of MTD/RD and objective tumor response

13. Study Procedures & Methodology

Management Pathway: Standard early-phase oncology care pathway **Education Protocol (HCP):** Investigator meetings detailing DLT definitions, combination therapy risks, and strict biopsy/PK sampling procedures **Education Protocol (Patient):** Informed consent detailing sequential tumor biopsy requirements and potential immune-related adverse events **Quality Audit Mechanism:** Routine sponsor audits and independent ethical committee oversight

14. Observation & Follow-Up Schedule

Total Duration: Estimated 7+ years (from start to primary completion) **Onsite Visit Cadence:** Frequent monitoring during Cycle 1 (e.g., weekly) for DLT assessment, transitioning to standard infusion cycles for subsequent treatment phases. **Remote Monitoring Frequency:** Routine survival check-ins post-treatment **Checklist Review Interval:** Tumor imaging typically every 8-12 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				